

Homework — 1.2.3 Software Development — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] — Correct order:

Stage	Order
Testing	4
Analysis (requirements)	1
Evaluation / maintenance	5
Design	2
Development (coding)	3

All five correctly placed = 2 marks; three or four correctly placed = 1 mark.

2. [3] — 1 mark per row (correct verdict; for false statements the correction is required):

Statement	True / False	Correction
XP developers always work alone	False	XP uses pair programming — two developers share one workstation, one coding while the other reviews
Waterfall client involvement mainly at start and end	True	—
No point testing erroneous data	False	Erroneous data is included in a test plan to check that invalid input is rejected gracefully / handled with an error message rather than crashing

3. [3] — 2 correct cells per mark:

Type	Example	What it checks
Normal (valid)	(a) e.g. 57 (any value 1–99)	(b) That typical valid input is accepted and processed correctly
Boundary	(c) 0 or 100 (accept just-outside values –1/101)	(d) That the limits of the range are handled correctly — errors most often occur at the edges (e.g. > used instead of >=)
Erroneous	(e) e.g. 150, –20 or "abc"	(f) That invalid input is rejected / produces a suitable error message without crashing

4. [2] — 2 correct rows per mark: Waterfall = **B**; Extreme programming = **D**; Rapid application development = **A**; Spiral = **C**. (*E and F are distractors: compilation and white-box module testing.*)

5.

(a) [3] — 1 mark per point, max 3: Waterfall fixes the requirements at the start and each stage must be completed before the next begins, so a change forces the project back to an earlier stage — expensive/slow (1). Agile works in short iterations, delivering working software each cycle, so requirements can be re-prioritised/added between iterations (1). The client is involved throughout and gives feedback on each iteration, so the final product better matches their (changing) needs (1). *(Accept the converse points about waterfall.)*

(b) [1] — Any one: code can be less efficient/less well structured because speed is prioritised; poorer documentation; needs continual user availability; unsuited to safety-critical systems where full up-front design is needed.

Part B (6 marks)

6. [2] — Any two (1 each): compiler translates all at once before running, interpreter translates and executes line by line at run time; compiler produces a standalone executable, interpreter produces none; compiled code runs faster; interpreted code needs the source + interpreter present each run; interpreter stops at the first error, compiler reports errors at the end. Max 2.

7. [2] — Washing machine → **embedded OS**, because it performs one dedicated task with limited resources, is stored in ROM and must be reliable/efficient (1). Server cluster → **distributed OS**, because it runs across multiple networked computers as one system, sharing resources/workload (1).

8. [2] — RAM is volatile — loses its contents when power is removed — and holds the programs and data currently in use (1). ROM is non-volatile — keeps its contents without power — and stores firmware such as the BIOS/boot program (1).

Total: 14 + 6 = 20